

Political Economy · Ideias

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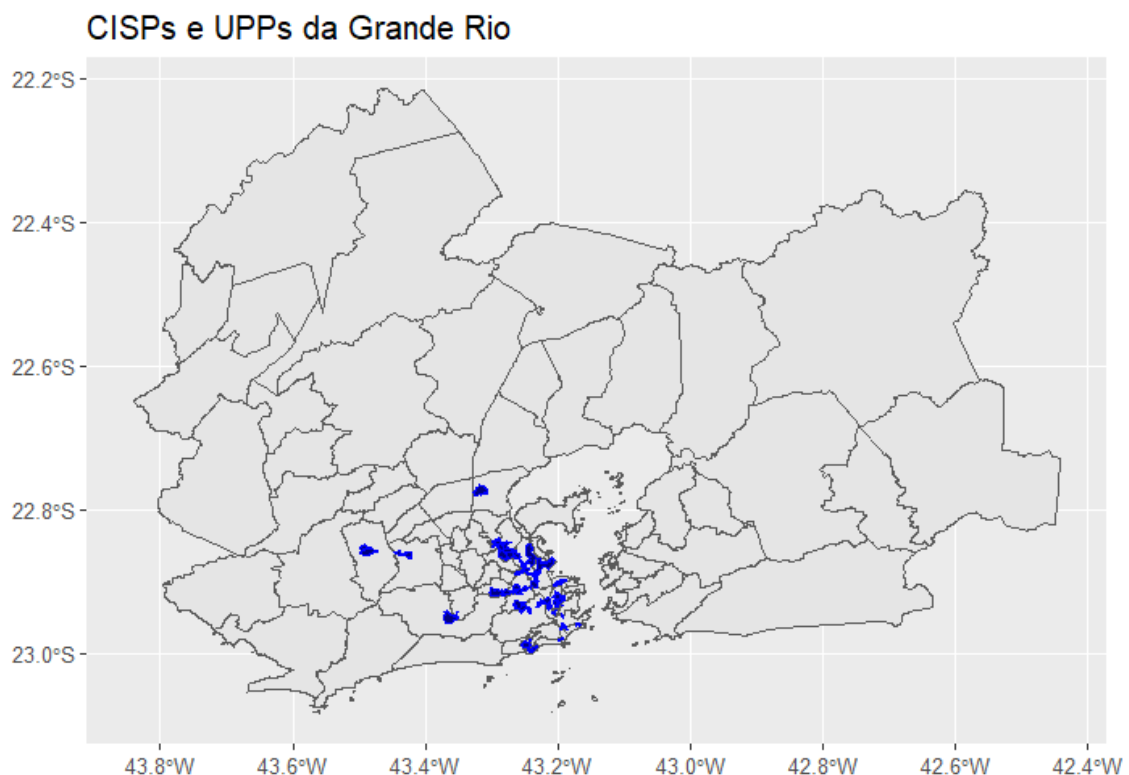


1 Dados

Daqui a pouco vou colocar um dicionário dos arquivos no [repositório](#).

1.1 Violência

Menores níveis hierárquicos: CISP¹ e UPP.



Dá pra ver os plots de outros níveis rodando o `alpha.R`.

Tem todos os tipos de crimes possíveis por CISP e UPP, série mensal (tem o dicionário de variáveis no [./docs](#) do repositório). Coloquei o dataset geral e o de feminicídio no [./data](#) já.

¹Do [MPRJ](#): “As Circunscrições Integradas de Segurança Pública - CISP, correspondem às áreas territoriais de atuação e responsabilidade conjunta das delegacias distritais da Secretaria de Estado de Polícia Civil (SEPOL) e das companhias integradas da Secretaria de Estado de Polícia Militar (SEPM).”

1.2 Tiroteios + mapa de controle da cidade

Instituto Fogo Cruzado + GENI/UFF.

Tô tentando descobrir como extrair o mapa (se tiver como; se der pra pensar num experimento natural, extração manual não seria problema). Coloquei dados que eu tô usando de tiroteio no Rio de 2019-2020 por enquanto, mas dá pra puxar pra parte da Grande Rio (e algumas outras regiões metropolitanas) bem fácil.

1.3 Outros

Não sei se vai ser útil, mas coloquei dados de quais cidades tomaram quais medidas de quarentena no Brasil e dados de COVID do DATASUS.

Tem o shp da prefeitura do Rio também (inclui algumas outras cidades, a maioria na Baixada).

2 Metodologia

- Dá pra agrupar os dados de violência a nível de CISP ou UPP pra regressão.
- Literatura (ver apêndice): pra controlar endogeneidade de presença policial, usam coincidência de partido de prefeito-governador, painel GMM (lagged vbs; também pra inércia do crime); pra erro de mensuração dos crimes reportados, modelam o erro como aleatório; pra inversão de causalidade, uso de transporte.

Algumas variáveis independentes (tô vendo o que eu consigo a níveis mais baixos que município):

- IDH, medidas de educação, renda, analfabetismo, pobreza extrema, desemprego, raça, população (+ população entre 15-24 anos), domicílios liderados por mulheres, taxa de urbanização, taxa de crescimento do produto, idade, prior offenses

Outras que eu pensei:

- Pra população entre 15-24 (ou até 29), podia colocar gênero também;
- Experimentos naturais: instalação das UPP $\rightsquigarrow$ ver se crime deslocou (e.g. pra Niterói); controlar pra impedimentos geográficos ou da malha rodoviária pra tentar extrair variação exógena
- Hipótese: várias facções em uma só área na regressão deviam gerar mais conflitos organicamente. Se só duas facções: ver se existe informação sobre acordos de controle compartilhado (e.g. CV + milícia).
- Algumas IVs pra desigualdade poderiam ser altura + elevação das construções (casas na favela têm menos andares, ficam mais em morros), arranjo das ruas (não costumam ter um grid planejado na favela).
 - Parece overkill, mas talvez sirva de alguma coisa se a gente conseguir dados de desigualdade pra áreas agregadas pelo CPS etc.

3 Apêndice

Alguns artigos sobre o assunto. As tabelas só servem pra mostrar quais foram as variáveis independentes e alguns resultados sobre os coeficientes delas.

3.1 *The Socioeconomic Determinants of Crime in Brazil*: the role of spatial spillovers and heterogeneity (2020)

TABLE 2

Factorial loadings and commonality

Variables	Factorial loadings		Uniqueness
	F1	F2	
HDI	0.9536	-0.1997	0.0508
Higher_education	0.8919	-0.0098	0.2045
Income_pc	0.8884	-0.2254	0.1600
Illiterate_pop	-0.8852	0.3347	0.1044
Extremely_poor	-0.8575	0.3023	0.1734
Primary_school	-0.8537	-0.0295	0.2703
Unemployed_pop	-0.1140	0.8439	0.2747
Women_heads	0.0909	0.7770	0.3880
Non-white_pop	-0.5116	0.7062	0.2396
Population_15-24	-0.4377	0.5725	0.4806

Source: research data.

TABLE 5

OLS Regression.

Independent variables	Coefficient	Standard deviation	T-statistics	P-value
Constant	-0.7841**	0.1170	-6.7033	0.0000
Economic and Educational Factor	-0.0520**	0.0081	-6.4561	0.0000
Social Disorganization Factor	0.0592**	0.0092	6.4322	0.0000
Demographic Density	0.0163	0.0141	1.1544	0.2484
Gini Index	0.4339*	0.1749	2.4800	0.0132
Population	0.4363***	0.0214	20.3788	0.0000
Jarque-Bera	119.733**			0.0000
Koenker-Basset	537.527**			0.0000

Source: research data.

Note: ** Significant at 1%; * Significant at 5%.

Nível: municipal.

Usaram factor analysis em X pra lidar com multicolineariedade, tratam de spillovers com um AR espacial (SAR).

3.2 *The Effects of Inequality on Crime: A Cross-Sectional Analysis (2017)*

VARIABLES	(1) GINI Index	(2) GINI Index	(3) GINI Index	(4) GINI Index
Police Inquiry Rate 2010 (%)	1.555*** (0.565)	1.532*** (0.551)	1.161** (0.555)	1.279** (0.535)
Robbery Rate 2009	9.41e-05*** (1.06e-05)	9.11e-05*** (9.15e-06)		
Logged GDP per capita	0.00291 (0.00552)		0.0172*** (0.00562)	
Urbanization Rate (%)	-0.000255 (0.000171)		0.000165 (0.000178)	
High School Dropout Rate(%)	0.000415 (0.000503)		0.000528 (0.000535)	
Formal Employment Rate(%)	-0.000665 (0.0194)		-0.0137 (0.0218)	
Constant	0.418*** (0.0504)	0.427*** (0.00596)	0.268*** (0.0508)	0.447*** (0.00558)
Observations	645	645	645	645
F-Statistic	18.18***	51.51***	4.24***	5.71***
R-squared	0.163	0.158	0.034	0.007

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Nível: municipal.

Instrumentam efeito da polícia com alinhamento do prefeito com o PSDB (partido do estado).

3.3 *Inequality and Criminality Revisited*: further evidence from Brazil (2004)

Table 4.

Econometric Model for Intentional Homicide: Panel Data GMM System Estimator*

Independent Variables	(1)	(2)	(3)	(4)
Mortp_1**	–	0.617 (0,000)	–	0.932 (0,000)
Poverty	-0.235 (0.022)	-0.127 (0.058)	-0.620 (0.000)	-0.154 (0.007)
Gini	0.803 (0.003)	0.667 (0.012)	0.672 (0.000)	0.780 (0.024)
Urbanization	0.860 (0,064)	0.787 (0.056)	0.720 (0,000)	0.408 (0.101)
Unemployment	0.202 (0.097)	0.071 (0.109)	0.364 (0.000)	0.050 (0.050)
Public Security	-0.322 (0,064)	-0,078 (0,000)	-0.373 (0,000)	-0,048 (0,059)
SchoolMale15	0.948 (0.002)	0.071 (0.006)	–	–
Constant	-4.561 (0.0080)	1.473 (0.001)	5.586 (0.000)	1.007 (0.007)
Serial Correlation AR(1)	-1.300 (0.002)	-3.100 (0.002)	-1.300 (0.002)	-4.640 (0.002)
Serial Correlation AR(2)	0.062 (0.952)	0.300 (0.765)	0.005 (0.982)	-0.180 (0.858)
Sargan Test	91.71 (0.992)	121.90 (0.587)	97.52 (0.982)	143.09 (0.459)
Observations	309	263	309	263

The values in parentheses are the p-values of the variables. ** Mortp = terminology for intentional homicide,. Mortp_1 = First lag of Mortp.

Nível: estadual.

Polícia controlada com GMM, painel. Usam Granger etc. pra ver se tem inversão (não acham).

3.4 Inequality and Violent Crime (2002)

Nível: países.

“To account for criminal inertia, we need to work with a dynamic, lagged dependent econometric model.”

[...]

“We address the problem of joint endogeneity by employing an instrumental variable procedure applied to dynamic models of panel data. This is the GMM estimator that uses the dynamic properties of the data to generate proper instrumental variables.” (**lagged vbs**)

[...]

“To control for measurement error, we model it as either random noise or a combination of an unobserved country-specific effect and random noise.”

TABLE 4
BASIC ECONOMIC MODEL (Generalized Method of Moments Estimation)

	HOMICIDE RATE			ROBBERY RATE		
	(1)	(2)	(3)	(4)	(5)	(6)
Regression specification	Levels	Levels*	Levels and differences	Levels	Levels*	Levels and differences
Lagged dependent variable	.8957 (46.2310)	.9282 (114.5404)	.8137 (25.4593)	.7254 (12.3614)	.7528 (23.1968)	.7222 (50.0311)
Income inequality: Gini coefficient	.0069 (3.9761)	.0032 (2.3130)	.0155 (7.0490)	.0331 (3.5354)	.0223 (4.8849)	.0307 (11.8691)
Growth rate: % annual change in real gross domestic product	-1.9270 (-2.9066)	-3.3952 (-6.5945)	-4.2835 (-4.9471)	-8.4505 (-7.2343)	-7.1754 (-9.3441)	-11.1536 (-18.1176)
Average income: log of gross national product per capita in U.S. dollars	-.0570 (.0187)	-.0396 (-4.0141)	.0151 (.8876)	-.0541 (-.7641)	-.0923 (-2.9946)	-.0287 (-2.0664)
Urbanization: % urban population	-.0032 (-4.9258)	-.0023 (-5.9852)	-.0019 (-.8959)	.0078 (4.0733)	.0106 (6.9098)	.0053 (6.0005)
Educational attainment: average years of education for adults	-.0090 (-1.1584)	-.0153 (-2.7694)	-.0300 (-3.4280)	.0901 (3.9416)	.0634 (4.2398)	.0382 (6.5551)
Intercept	.7935 (5.3834)	.7584 (9.7501)		-.5486 (-.8663)	.0669 (.2277)	
Number of countries	39	27	27	37	29	29
N	106	91	91	94	85	85
Specification tests, <i>P</i> -values:						
Sargan test	.651	.581	.958	.531	.314	.430
Serial correlation:						
First order	.683	.873	.048	.035	.103	.082
Second order	.239	.498	.240	.147	.225	.879

SOURCES.—Authors' calculations. For details on definitions and sources of variables, see Appendix A, Table A1. Homicides data source: World Health Organization's mortality statistics; robbery data source: United Nations' world crime surveys.

NOTE.—The *t*-statistics are in parentheses.

* The sample is restricted to the countries that have at least three consecutive observations.

TABLE 5
ALTERNATIVE INEQUALITY MEASURES (Generalized Method of Moments Levels Estimation)

DEPENDENT VARIABLE (in logs)	HOMICIDE RATE			ROBBERY RATE		
	(1)	(2)	(3)	(4)	(5)	(6)
Lagged dependent variable	.8780 (48.8267)	.8688 (56.5593)	.9268 (58.2480)	.7818 (18.7384)	.7784 (25.1457)	.8698 (18.9108)
Growth rate: % annual change in real gross domestic product	-3.2533 (-3.8806)	-3.1665 (-4.2468)	-.9411 (-1.2659)	-4.1422 (-3.6406)	-4.7806 (-4.1412)	-6.9070 (-2.3018)
Average income: log of gross national product per capita in U.S. dollars	-.0666 (-4.2536)	-.0761 (-4.1829)	-.0590 (-2.8788)	-.0666 (-1.8281)	-.1135 (-3.1697)	.0373 (.6770)
Urbanization: % urban population	-.0019 (-2.1161)	-.0023 (-2.4541)	-.0031 (-2.5615)	.0078 (3.6231)	.0095 (4.3303)	.0028 (.9051)
Educational attainment: average years of education for adults	-.0205 (-3.1268)	-.0151 (-2.5024)	-.0083 (-.7124)	.0568 (2.6936)	.0699 (2.9841)	.0632 (1.9850)
Intercept	1.0556 (8.5890)	.7916 (5.6564)	.9411 (5.1455)	.4020 (1.5376)	.0463 (.1752)	-.7550 (-1.2889)
Ratio of the first to the fifth quintile	.0152 (3.7918)			.0426 (4.0275)		
Income polarization: log of income polarization index		.0019 (4.9967)			.0037 (3.6931)	
Educational inequality: standard deviation of schooling years			.1036 (.8801)			.8976 (2.0666)
Number of countries	39	39	37	36	36	35
N	96	96	103	86	86	91
Specification tests: <i>P</i> -values:						
Sargan test	.533	.789	.512	.168	.201	.180
Serial correlation:						
First order	.662	.742	.829	.145	.078	.023
Second order	.272	.272	.174	.283	.136	.359

SOURCES.—Authors' calculations. For details on definitions and sources of variables, see Appendix A, Table A1. Homicides data source: World Health Organization's mortality statistics; robbery data source: United Nations' world crime surveys.

NOTE.—The *t*-statistics are in parentheses.

TABLE 6
ADDITIONAL CONTROL VARIABLES (Generalized Method of Moments Levels Estimation)

DEPENDENT VARIABLE (in logs)	HOMICIDE RATE				ROBBERY RATE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Lagged dependent variable	.8792 (57.6962)	.8987 (62.3507)	.9040 (45.7224)	.9106 (42.0085)	.8195 (15.2561)	.7358 (12.9489)	.7414 (12.2210)	.7849 (16.3984)
Income inequality: Gini coefficient	.0069 (2.7692)	.0055 (3.4744)	.0049 (2.1574)	.0073 (3.8160)	.0275 (3.3464)	.0307 (3.3330)	.0228 (2.4784)	.0226 (3.8775)
Growth rate: % annual change in real gross domestic product	-1.3845 (-2.8885)	-1.0327 (-1.3381)	-1.8185 (-2.4141)	-9918 (-1.1735)	-7.8263 (-6.9305)	-8.6606 (-6.2523)	-7.1703 (-4.9902)	-8.1747 (-6.0847)
Average income: log of gross national product per capita in U.S. dollars	-.0335 (-1.2479)	-.0689 (-6.2284)	-.0432 (-2.4857)	-.0448 (-2.9894)	.0115 (.1548)	-.0332 (-.5184)	-.0477 (-.7148)	-.0865 (-1.6034)
Urbanization: % urban population	-.0033 (-5.6910)	-.0029 (-4.7985)	-.0033 (-5.0455)	-.0031 (-4.9780)	.0047 (2.4969)	.0065 (2.9605)	.0045 (2.3226)	.0078 (4.6855)
Educational attainment: average years of education for adults	-.0203 (-2.1061)	-.0042 (-.6974)	-.0103 (-1.2640)	-.0029 (-.4580)	.0453 (3.0588)	.0832 (3.7235)	.1081 (4.5435)	.0707 (4.6855)
Intercept	.6323 (2.5482)	.8846 (8.5859)	.7445 (4.7312)	.7186 (6.5509)	-.6947 (-1.0287)	-.6327 (-1.0006)	-.2877 (-.4495)	.0029 (.0089)
Ethnolinguistic fractionalization	.1706 (2.1826)				-.1092 (-.7786)			
Police per 100,000 population		-.0001 (-.9153)				.0004 (1.5252)		
Latin America: dummy variable			.0493 (.9198)				.3486 (2.7248)	
Young male population share: 15–29-year-olds as % of total population				-.0103 (-1.5412)				.0039 (.1930)
Number of countries	34	35	39	39	31	36	37	37
Number of observations	96	97	106	106	83	92	94	94
Specification tests: <i>P</i> -values:								
Sargan test	.674	.533	.471	.536	.554	.575	.349	.539
Serial correlation:								
First order	.486	.702	.732	.782	.060	.025	.044	.043
Second order	.123	.240	.279	.244	.153	.120	.175	.194

SOURCES.—Authors' calculations. For details on definitions and sources of variables, see Appendix A, Table A1. Homicides data source: World Health Organization's mortality statistics; robbery data source: United Nations' world crime surveys.

NOTE.—The *t*-statistics are in parentheses.

3.5 *Crime and Inequality*: Reverse Causality? (2007)

Níveis: Standard Metropolitan Areas.

Table 2: Mobility and Property Crime 2006									
Dependent Variable: % Population Moving Within County 2005/2006 Per Income Bracket									
	LESS10K		10K15K		15K25K		25K35K		
	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	
C	2.091	0.404 ***	0.536	0.353	0.108	0.300	-0.273	0.294	
LOG(PCRIME)	0.005	0.016	0.022	0.014	0.025	0.012 **	0.025	0.011 **	
LOG(ME)	-0.416	0.118 ***	-0.175	0.103 *	-0.193	0.088 **	-0.149	0.086 *	
LOG(AGE)	0.008	0.030	0.040	0.026	0.005	0.022	-0.007	0.022	
LOG(RACE)	0.009	0.005	0.009	0.005 *	0.004	0.004	-0.002	0.004	
LOG(POPULATION)	-0.007	0.002 ***	-0.006	0.002 ***	-0.003	0.002	-0.002	0.002	
FAMILY_PC	-0.244	0.123 **	-0.019	0.108	0.237	0.091 **	0.234	0.090 **	
FEMALEHH_PC	0.065	0.163	-0.217	0.143	-0.191	0.121	-0.020	0.119	
LESSHS_PC	-0.498	0.129 ***	-0.228	0.113 **	-0.324	0.096 ***	-0.305	0.094 ***	
HSD_PC	-0.314	0.081 ***	-0.180	0.070 **	-0.178	0.060 ***	-0.175	0.059 ***	
LOG(MEDROVEALONE)	0.266	0.114 **	0.117	0.100	0.170	0.085 **	0.161	0.083 *	
LOG(SOMECOLL_PC)	-0.496	0.123 ***	0.024	0.107	0.020	0.091	0.073	0.089	
LOG(WCMOVERS)	1.656	0.173 ***	1.394	0.151 ***	0.983	0.128 ***	0.788	0.126 ***	

...

Table 5: Means of Transportation to Work and Violent Crime								
	(I)		(II)		(III)		(IV)	
Dependent Variable	LOG(MECARPOOL)		LOG(MECARPOOL)		LOG(MECARPOOL)- LOG(MECARPOOL2004)		LOG(MECARPOOL)- LOG(MECARPOOL2004)	
	Coef.	P-Value	Coef.	P-Value	Coef.	P-Value	Coef.	P-Value
C	2.707	0.000	1.312	0.140	-5.109	0.022	-5.456	0.035
LOG(VCRIME)	-0.065	0.002	-0.076	0.019	-	-	-	-
LOG(VCRIME)- LOG(VCRIME2004)	-	-	-	-	-0.189	0.095	-0.156	0.225
LOG(ME)	2.174	0.000	1.931	0.000	-1.091	0.146	-0.368	0.538
LOG(AGE)	-0.165	0.011	-0.022	0.795	0.147	0.295	-0.001	0.997
LOG(RACE)	-0.033	0.005	-0.035	0.016	0.068	0.042	0.056	0.136
LOG(POPULATION)	-0.006	0.278	-0.005	0.418	0.041	0.206	0.076	0.058
FEMALEHH_PC	-0.410	0.286	0.304	0.549	-1.396	0.064	-0.997	0.317
FAMILY_PC	-0.262	0.239	-0.553	0.061	1.165	0.027	1.260	0.023
HSD_PC	0.046	0.789	-0.109	..	0.838	0.018	0.691	0.097
LOG(MEDROVEALONE)	-1.334	0.000	-0.976	0.002	1.417	0.058	0.791	0.179
CARPOOLERS_PC	0.285	0.492	-	-	-2.794	0.063	-	-
PUBLICTRANSP_PC	0.647	0.066	-	-	4.046	0.197	-	-
LOG(CARPOOLERS_PC/ PUBLICTRANSP_PC)	-0.014	0.250	-	-	0.090	0.093	-	-
MINUTES_CP	-	-	-0.011	0.371	-	-	-0.048	0.289
MINUTES_PT	-	-	0.008	0.345	-	-	0.025	0.445
LOG(MINUTES_CP/MIN UTES_PT)	-	-	0.328	0.321	-	-	1.044	0.405
R-squared	0.797		0.813		0.345		0.419	
N. Obs	151		88		56		42	

3.6 *Economic Inequality, Relative Deprivation, and Crime: An Individual-Level Examination*

Nível: individual.

“The main independent variable is relative deprivation. Several indices have been offered to measure relative deprivation, of which the most popular are currently the Yitzhaki index and income rank. The former’s theoretical basis is that what matters in income comparisons is the magnitude of differences between individuals; the index takes this into account by calculating the average difference between an individual’s income and those of all other individuals with higher incomes. The income rank index, by contrast, is based on the idea that people lack knowledge of precisely how much others earn, and so what matters is their ordinal position, or rank, within a given group. Following this, and given that the income rank measure is correlated more strongly than the Yitzhaki index with negative outcomes, we operationalized relative deprivation using this measure.”

Variable	Property crimes			Violent crimes		
	HR	RSE	CI	HR	RSE	CI
Lag relative deprivation	1.58***	0.07	1.44, 1.74	1.48***	0.05	1.38, 1.58
Log income	0.97***	0.01	0.97, 0.98	0.99***	0.01	0.98, 0.99
Log spousal income	0.98***	0.01	0.97, 0.98	0.99***	0.01	0.98, 0.99
Number of prior offenses	1.78***	0.02	1.75, 1.82	1.54***	0.01	1.52, 1.56
Male (1 = Yes)	4.51***	0.16	4.21, 4.83	9.53***	0.31	8.93, 10.16
Age	0.97***	0.01	0.96, 0.97	0.97***	0.01	0.97, 0.97
Education years	0.87***	0.01	0.86, 0.88	0.86***	0.01	0.85, 0.87
Jewish (1 = Yes)	0.92	0.04	0.84, 1.01	0.68***	0.02	0.65, 0.72
Married (1 = Yes)	0.55***	0.02	0.51, 0.60	0.89***	0.03	0.83, 0.94
Born in Israel (1 = Yes)	0.75***	0.03	0.70, 0.81	0.56***	0.01	0.53, 0.59

3.7 Outros

3.7.1 *Inequality and Crime* (2000)

Teoria econômica do crime explicaria crime à propriedade, mas não crimes violentos.

3.7.2 *Inequality and Violent Crime: Evidence from Data on Robbery and Violent Theft* (2005)

Efeito de desigualdade no crime é dependente de FE de cultura.